

CABLETWIN -- DAY 3 INDUSTRY PITCH

FORMAT: 4:00 pitch - 4:00 Q&A - maximum 0:30 demo

ROOM: Industry - Room 206

STATUS: FINAL -- frozen for the 14:00 session.

Live-demo wording on slide 4 still requires runtime ACCEPT (decision 13:30);
the pre-accept fallback below is the default.

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ONE COMMUNICATION JOB

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The jury should select CableTwin and connect one Tunisian cable-plant design partner because it demonstrates a bounded, human-governed decision layer that compares recovery options before production changes.

Core promise:

Decision clarity before production changes.

Closing line:

The ML retrieves the past. The twin compares futures. Hard rules block the impossible. Humans and the laboratory authorize.

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EXACT FOUR-MINUTE SCRIPT

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SLIDE 1 -- 0:00-0:20 -- HOOK

Good afternoon. In our synthetic Tunisian cable-factory scenario, grid power disappears for 58 minutes while a medium-voltage cable order is in production. Electricity returns; certainty does not. CableTwin compares recovery options before the plant commits equipment, quality resources, people and the customer schedule.

SLIDE 2 -- 0:20-0:50 -- PROBLEM

Why is this difficult? The line sees the stop. Energy sees a restart envelope. Quality sees incomplete process history. Logistics sees a truck at 06:30. One 24-kilometre order now has three truths: drums one to six remain outside the incident, drum seven is entirely on hold, and drum eight is not started and must be replanned. Restarting the line is not the same as recovering the operation.

SLIDE 3 -- 0:50-1:20 -- VALUE + TECHNICAL LOGIC

CableTwin is a local, read-only decision layer. Explainable ML retrieves comparable synthetic cases and shows distance, differences and out-of-distribution warning. The deterministic twin tests recovery futures across energy, process, material, people, quality and delivery. Hard rules reject impossible sequences. CableTwin never commands a PLC and never releases cable: Quality, the laboratory and accountable humans keep authority.

SLIDE 4 -- 1:20-1:50 -- DEMO

LIVE WORDING -- USE ONLY AFTER RUNTIME ACCEPT:

Here is the same scenario in Jury mode. I inject the 58-minute outage: drum seven stays on hold and the system retrieves comparable cases. It rejects restart-all and emergency production on hard rules. I reduce available restart power: no plan is feasible. I restore the operator-entered limit: staged Plan C returns under visible assumptions. The laboratory gate remains closed. Same engine, same rules, faster replay--not a scripted shortcut.

PRE-ACCEPT FALLBACK (DEFAULT):

This is the thirty-second acceptance contract now under review: incident, hold state, comparable cases, two branches rejected on hard rules, a live no-feasible-plan stress test, then a staged recovery that still leaves the laboratory gate closed. Until the same-engine and browser gates pass, we present it as a specification, not as a live capability.

SLIDE 5 -- 1:50-2:15 -- BUSINESS MODEL

We enter commercially in four disciplined steps: a paid data-and-decision diagnostic, a read-only shadow pilot, a license per line or site with configuration and support, then replication only after measured pilot evidence. Pricing and ROI are pilot questions. We sell decision speed, constraint visibility and traceability--not guaranteed savings.

SLIDE 6 -- 2:15-2:40 -- GO-TO-MARKET

The first pilot is one cable line for four to six weeks. We map the data contract and hard rules, replay historical incidents plus synthetic stress cases, compare CableTwin options with actual plant decisions, then go, revise or stop. Success measures are evaluation time, violations caught before action, stability, usability and traceability.

SLIDE 7 -- 2:40-3:05 -- COMPETITION

We complement existing systems; we replace none. SCADA and PLC control. MES executes and records. APS plans capacity. QMS and laboratories test and release product. CableTwin is the cross-domain, read-only decision layer that makes incident alternatives, hard constraints, provenance and human gates visible before the plant acts.

SLIDE 8 -- 3:05-3:25 -- TEAM + PROGRESS

I am Oussama Akir, a solo SUP'COM participant. During this hackathon I built the public CableTwin demonstrator, its bounded planner, local ML guidance, digital-twin views, verification and evidence package. I also kept the product's limits visible. The next team extension is not a fictional org chart; it is one real cable plant and its Quality authority.

SLIDE 9 -- 3:25-3:40 -- COMMERCIAL TRUTH

Today the honest numbers are zero signed pilots, zero revenue and zero ROI claims. What is proven is a public working product, nine planner checks, five ML checks, an exhaustive bounded verifier, and a local, offline, read-only posture. Synthetic evidence is not customer evidence.

SLIDE 10 -- 3:40-4:00 -- ASK + CLOSE

Our ask is precise: select CableTwin and connect one Tunisian cable-plant design partner for a read-only shadow pilot. A power interruption lasts 58 minutes; industrial uncertainty can last for hours. CableTwin retrieves the past, compares futures, blocks the impossible--and leaves authorization to humans and the laboratory.

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THIRTY-SECOND DEMO RUNBOOK

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Operator setup before the pitch:

- Open the accepted flagship route locally.
- Select JURY 30 S.
- Reset once.
- Confirm synthetic/non-affiliation banner is visible.
- Do not improvise labels, figures or routes.

0-05 s -- Incident

Action: trigger the 58-minute local outage.

Say: "Grid returns; decision uncertainty does not."

Must show: T1-T6 outside incident, T7 entirely HOLD, T8 not started.

05-11 s -- Memory

Action: advance to case retrieval.

Say: "ML retrieves comparable cases; it does not predict conformity."

Must show: distance, differences and OOD/provenance cues.

11-18 s -- Futures

Action: compare branches.

Say: "The twin rejects restart-all and emergency-production paths."

Must show: explicit hard-rule reasons.

18-24 s -- Live adversarial stress

Action: lower the operator-entered power limit.

Say: "Under this limit, no plan is feasible."
Must show: NO FEASIBLE PLAN; no silent fallback.

24-30 s -- Restore + authority
Action: restore the original limit and select staged Plan C.
Say: "Recovery becomes feasible; the laboratory gate stays closed."
Must show: T7 HOLD, T8 replanned, human/Quality gate.

Abort rule:
If the accepted route, reset, stress control or final state differs from the rehearsed evidence, stop the live interaction and use the static slide. Never explain a broken demo as a feature.

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FOUR-MINUTE ADVERSARIAL Q&A

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Q1 -- Is the solution genuinely functional or only a scenario?
A -- The current public demonstrator is functional, deterministic and read-only. The factory data and incident are synthetic. It is not an industrial qualification, a certified safety system or machine control.

Q2 -- Where is the AI if the cases are synthetic?
A -- The ML demonstrates a transparent retrieval-and-ranking mechanism: neighbors, distance, differences and OOD warning. A pilot must recalibrate and benchmark it on the plant's real historical incidents. It never predicts conformity.

Q3 -- How does CableTwin know which cable length is affected?
A -- It reconstructs an investigation envelope from position, time and encoded process assumptions. It does not locate a certain defect. The entire seventh drum remains HOLD until tests and Quality release.

Q4 -- Why say 24 km if only T7 and T8 are affected?
A -- Twenty-four kilometres is the full synthetic customer order. T1-T6 are outside the incident; T7 carries quality uncertainty; T8 carries the planning impact. We never claim to have saved 24 km.

Q5 -- Who is accountable if CableTwin recommends the wrong plan?
A -- CableTwin is consultative and read-only. Maintenance validates restart readiness, Energy confirms the power envelope, Quality holds or releases product, the laboratory provides evidence, Logistics accepts the partial shipment and the accountable plant manager signs the decision. CableTwin makes visible who must sign; it signs nothing.

Q6 -- Why can a 58-minute outage create hours of impact?
A -- The synthetic delay includes safe stop, inspection, ordered restart of auxiliaries, warm-up, stabilization, purge and controls. Every duration must be calibrated on the pilot line; 58 minutes implies no universal recovery duration.

Q7 -- Why do MES, APS, QMS, SCADA or Excel not solve this?
A -- They remain essential. CableTwin adds a bounded cross-domain what-if layer for one incident decision: energy, process, quality, people, material and delivery, with explicit provenance and human gates. It steps aside where an incumbent already solves the need.

Q8 -- What plant data does the pilot need?
A -- Line timestamps and states, energy meters, MES orders, quality events, material availability and operator decisions. The first pilot is read-only shadow mode with a planner-validated data contract.

Q9 -- How do you control cybersecurity risk?
A -- Local or edge deployment, network segmentation, least-privilege read-only inputs and an audit journal. No PLC command and no cloud dependency are required for the initial use case.

Q10 -- What is the business model and promised ROI?
A -- Paid diagnostic, bounded shadow pilot, then a per-line or per-site license with calibration and support. We promise no ROI before a measured baseline and pilot evidence.

Q11 -- How does it scale from a fictional line to many plants?

A -- The branch engine can be reusable, but constraints, data mappings, quality procedures and retrieval behavior are configured and validated per site. We do not claim a universal once-trained model.

Q12 -- Is the product useful only during Tunisian power cuts?

A -- No. The current Tunisian context makes the problem concrete. The same decision layer applies to site power limits, generator constraints, machine stops, material shortages and other industrial disruptions.

Q13 -- Yesterday's submission told a different demo story. Why the change?

A -- Yesterday proved the architecture on a scheduling incident: an exhaustive bounded verifier enumerated 17,856 candidates, 10,440 feasible schedules, and a traced human decision. Today I present the incident class the product exists for: one that crosses six domains. Same architecture, same limits, harder incident.

Q14 -- Why not run production on the backup generator?

A -- Sizing. The CCV process heaters and extruder drives dwarf a safety genset. In our scenario the generator keeps safety, control and cooling alive; production restart waits for the grid envelope.

Q15 -- What if the laboratory refuses to release drum seven?

A -- No surprise: Plan C never bet on T7. The 06:30 truck leaves with drums one to six, plus drum eight if the staged sequence holds. If the laboratory releases T7 it follows; if it refuses, a replacement drum is produced and the twin replans that order like any incident.

Q16 -- Why eight drums of three kilometres?

A -- The drum is the real decision unit: Quality holds a drum, the truck loads drums. The synthetic 8-by-3 structure gives the granularity that makes the decision visible: six outside, one held, one replanned.

Q17 -- Have you spoken to a cable plant?

A -- Not yet -- and that is precisely my ask. I bring a public product, written limits and a ready read-only pilot protocol. I ask for access rather than pretending to have it.

Q18 -- Is the retrieval running in the public product today?

A -- The retrieval mechanism is implemented and unit-tested; its flagship integration is in acceptance review. The public product ships the ranking advisor today: it scores the three planner-built routes and never creates or edits a schedule.

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CLAIM GUARDRAILS
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SAFE

- "Synthetic factory scenario."
- "Operator-entered outage window."
- "ML retrieves and ranks comparable cases."
- "The twin compares bounded futures."
- "Hard rules reject impossible plans."
- "Quality, the laboratory and humans retain authority."
- "Local, offline, read-only prototype."
- "Pilot targets to agree."
- "The exact verifier proves the encoded public demo (17,856 -> 10,440) -- never the flagship scenario."

DO NOT SAY

- "CableTwin predicts the outage."
- "CableTwin predicts or certifies cable conformity."
- "CableTwin saved 24 km."
- "CableTwin located the defect."
- "The generator powers production" -- the canonical scenario allows safety, control and cooling only.
- "Live STEG, PLC, SCADA, MES or plant integration."
- "Real factory data, customer pilot, revenue, ROI or savings."
- "Certified safety system."
- Never imply the exact verifier checked the flagship scenario.

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SOURCE NOTES

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Tunisian power-interruption context:

<https://www.tap.info.tn/fr/Portail-Titres-de-l-actualit%C3%A9/20404822-canicule-coupures>

<https://www.lapresse.tn/2026/07/17/alerte-steg-des-coupures-delectricite-possibles-ce-soir-de-17h-a-21h-dans-plu>

Human-in-loop digital-twin framing:

<https://www.nist.gov/publications/conceptual-architecture-digital-twins-human-loop-based-smart-manufacturing>

Industrial cable measurement context:

<https://sikora.net/en/products/xray8000nxt/>

Medium-voltage cable standard scope:

<https://webstore.iec.ch/en/publication/16382>

Incumbent APS comparison:

<https://www.siemens.com/en-gb/products/opcenter/advanced-planning-scheduling-aps/>

<https://www.3ds.com/products/delmia/ortems>

These sources support context and category framing. They do not validate the synthetic factory outcomes or any CableTwin plant-performance claim.